

Comparitive Performance Report for direct-reads-3-way-replica-balance-reads- off-pr2-run3 vs direct-reads-3-way-replica-balance-reads- on-split-off-pr2-run3 vs direct-reads-3-way-replica-balance-reads- on-split-on-pr2-run3

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Comparison summary for direct-reads-3-way-replica-balance-reads-off-pr2-run3 vs direct-reads-3-way-replica-balance-reads-on-split-off-pr2-run3 vs direct-reads-3-way-replica-balance-reads-on-split-on-pr2-run3

Sequential Read	direct_reads_3_way	direct_reads_3_way	direct_reads_3_way	direct_reads_3_way	direct_reads_3_way
4K	141819@0.4ms	233271@1.6	64%	232781@1.6	64%
8K	127922@2.0ms	224031@1.7	75%	220274@1.7	72%
16K	114767@3.3ms	207593@1.8	81%	204686@1.9	78%
32K	93685@4.1ms	160789@2.4	72%	157675@2.4	68%
64K	4953@5.1ms	7881@3.2	59%	7812@3.2	58%
128K	7946@6.3ms	11370@4.4	43%	11219@4.5	41%
256K	11444@8.8ms	14219@7.1	24%	14140@7.1	24%
384K	12152@16.6ms	14224@14.1	17%	14372@14.0	18%
512K	13052@20.6ms	13902@9.6	7%	13557@9.9	4%
768K	13366@30.1ms	13705@9.2	3%	13303@15.1	-0%
1024K	13806@29.2ms	12998@5.2	-6%	13569@29.7	-2%
2048K	12339@10.9ms	11855@11.3	-4%	13255@10.1	7%
4096K	11427@23.5ms	11160@48.1	-2%	12340@10.9	8%

Random Read	direct_reads_3_way	direct_reads_3_way	direct_reads_3_way	direct_reads_3_way	direct_reads_3_way
4K	161779@2.4ms	187678@2.0	16%	184917@2.1	14%
8K	154686@2.5ms	177754@2.2	15%	175190@2.2	13%
16K	145499@2.6ms	166889@1.9	15%	164603@1.9	13%
32K	123078@3.1ms	147620@2.6	20%	146199@2.6	19%
64K	6321@4.0ms	7716@3.3	22%	7731@2.7	22%
128K	9624@5.2ms	11373@4.4	18%	11254@3.0	17%
256K	12442@6.7ms	13476@2.5	8%	13458@6.2	8%
384K	13042@7.7ms	13795@3.6	6%	13735@4.6	5%
512K	13606@9.9ms	13827@4.8	2%	13035@5.1	-4%
768K	13595@7.4ms	13471@7.5	-1%	12950@3.9	-5%
1024K	13218@10.1ms	12979@5.2	-2%	13201@5.1	-0%
2048K	12222@11.0ms	11989@11.2	-2%	12909@10.4	6%
4096K	11446@70.4ms	11224@47.8	-2%	12349@9.5	8%

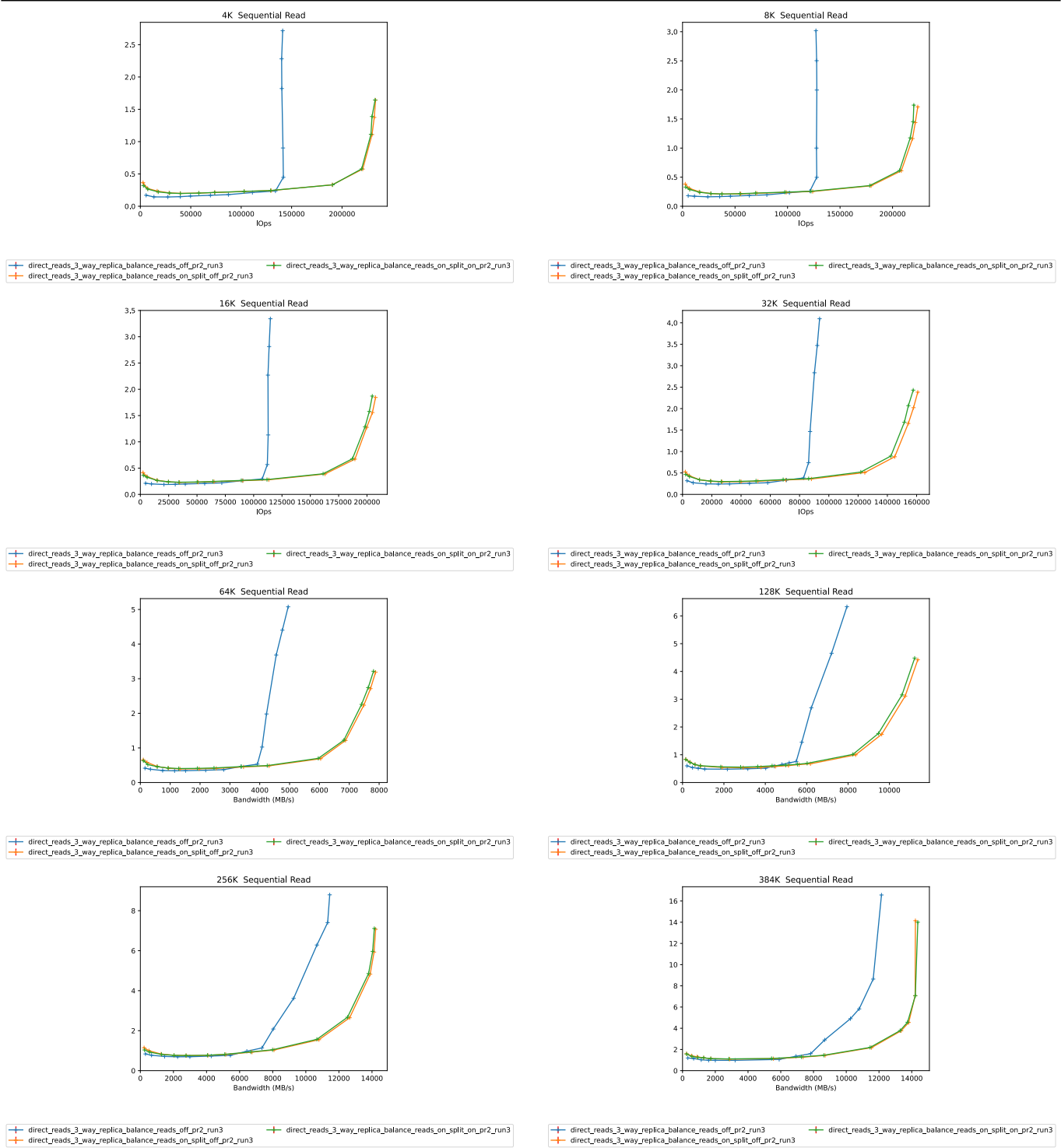
Random Read/Write	direct_reads_3_way	direct_reads_3_way	direct_reads_3_way	direct_reads_3_way	direct_reads_3_way
4K_70/30	88888@4.3ms	94162@4.1	6%	92832@4.1	4%
16K_70/30	81102@4.7ms	85626@4.5	6%	84642@4.5	4%

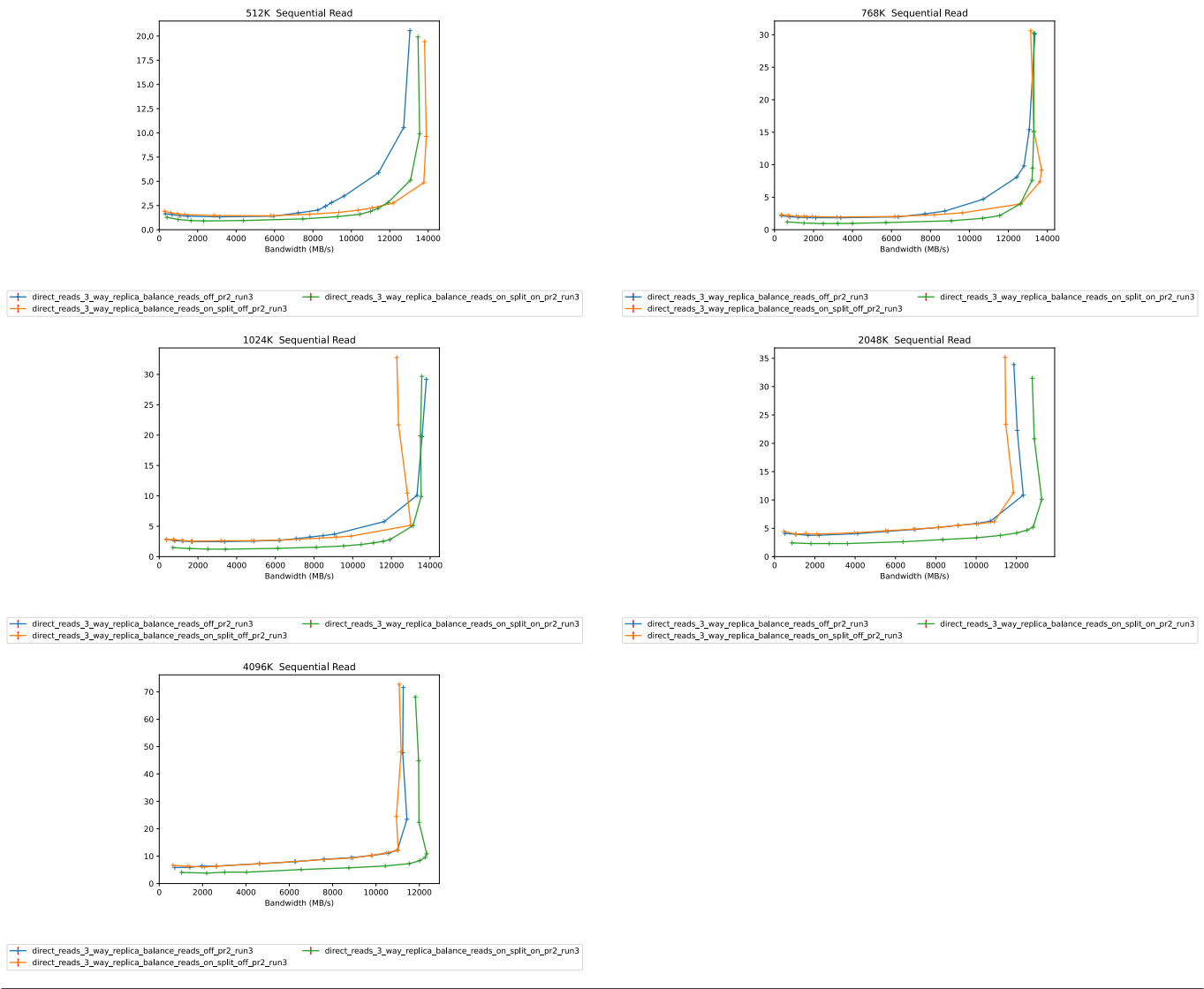
Comparison summary for direct-reads-3-way-replica-balance-reads-off-pr2-run3 vs
direct-reads-3-way-replica-balance-reads-on-split-off-pr2-run3 vs
direct-reads-3-way-replica-balance-reads-on-split-on-pr2-run3

Random					
Read/Write	direct_reads_3_way	direct_reads_3_way	direct_reads_3_way	direct_reads_3_way	direct_reads_3_way
64K_70/30	3940@6.4ms	4185@6.0	6%	4156@6.0	5%
64K_30/70	2793@7.5ms	2761@9.1	-1%	2761@7.6	-1%

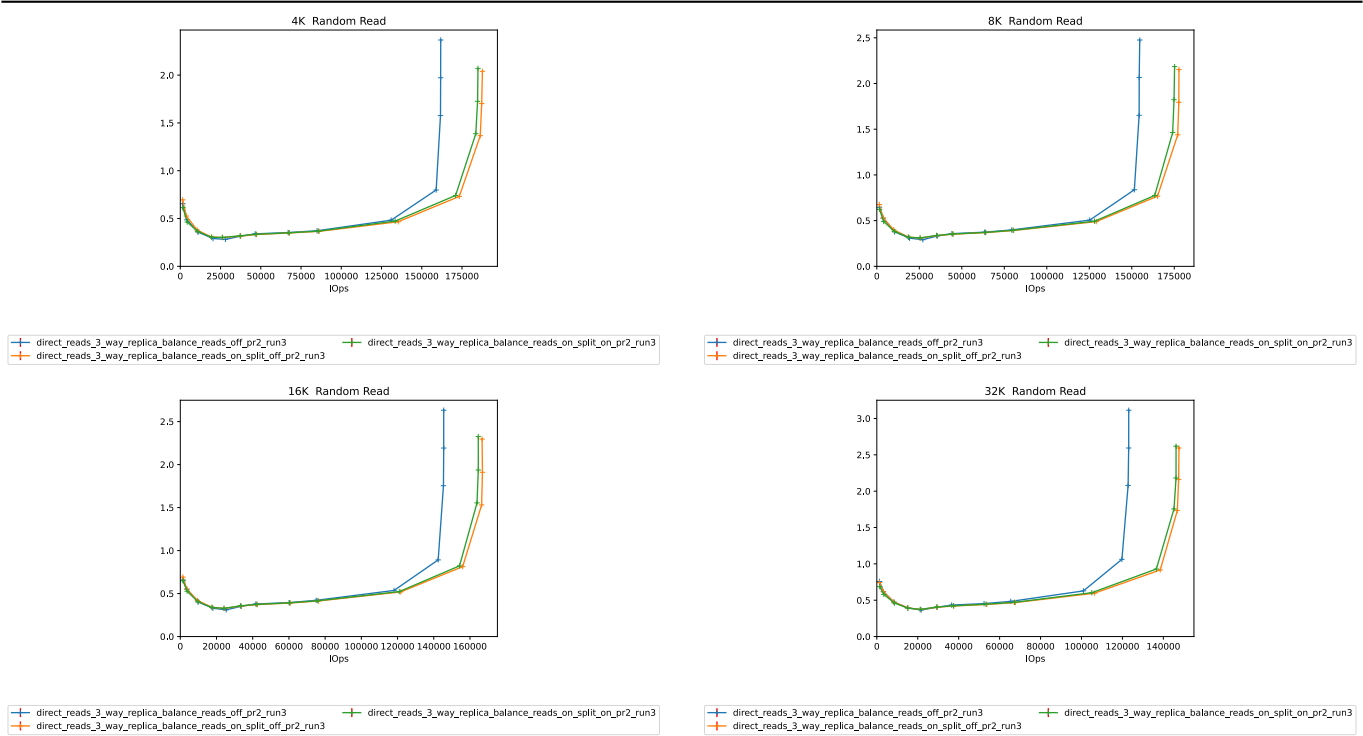
Response Curves

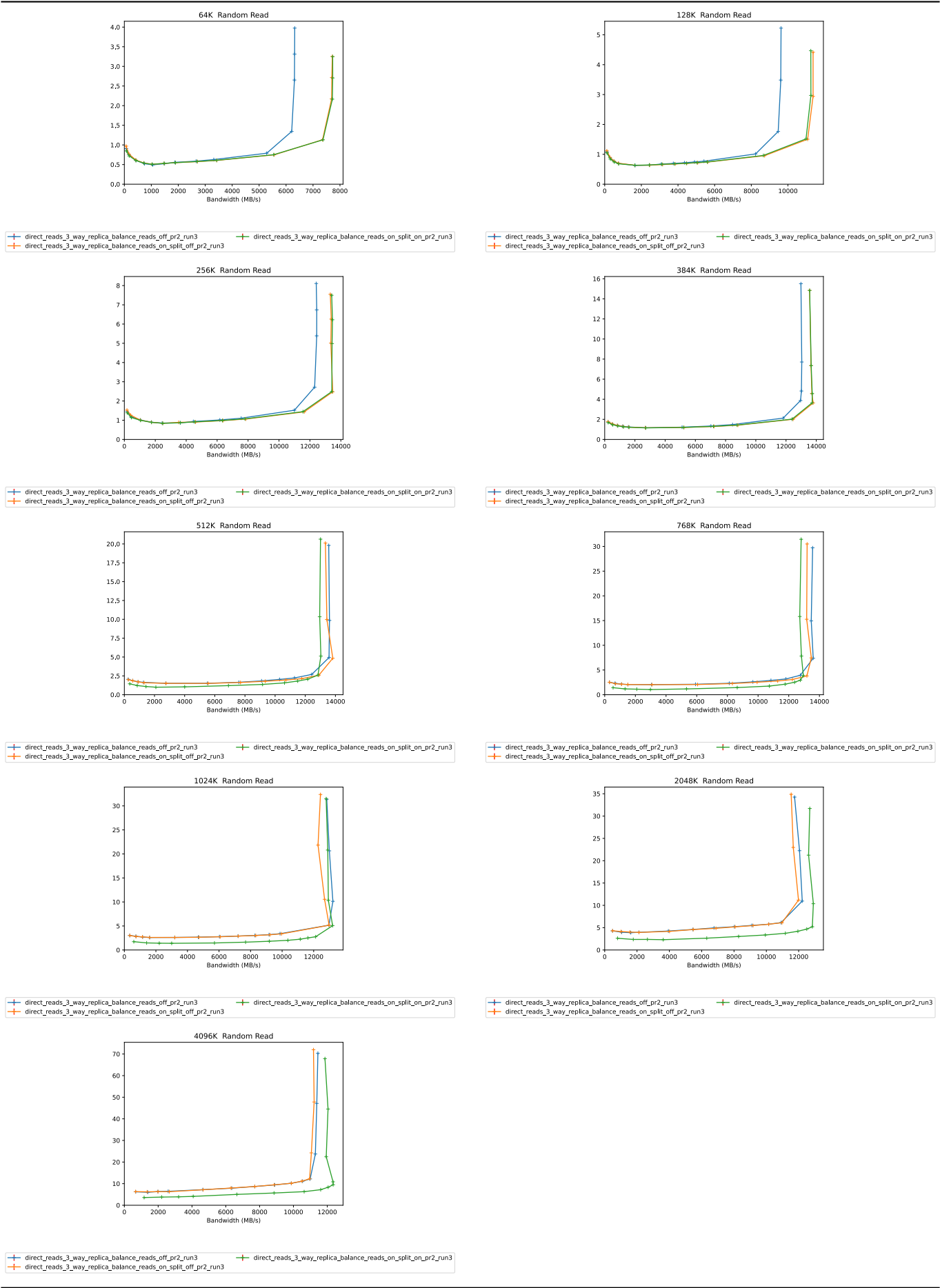
Sequential Read



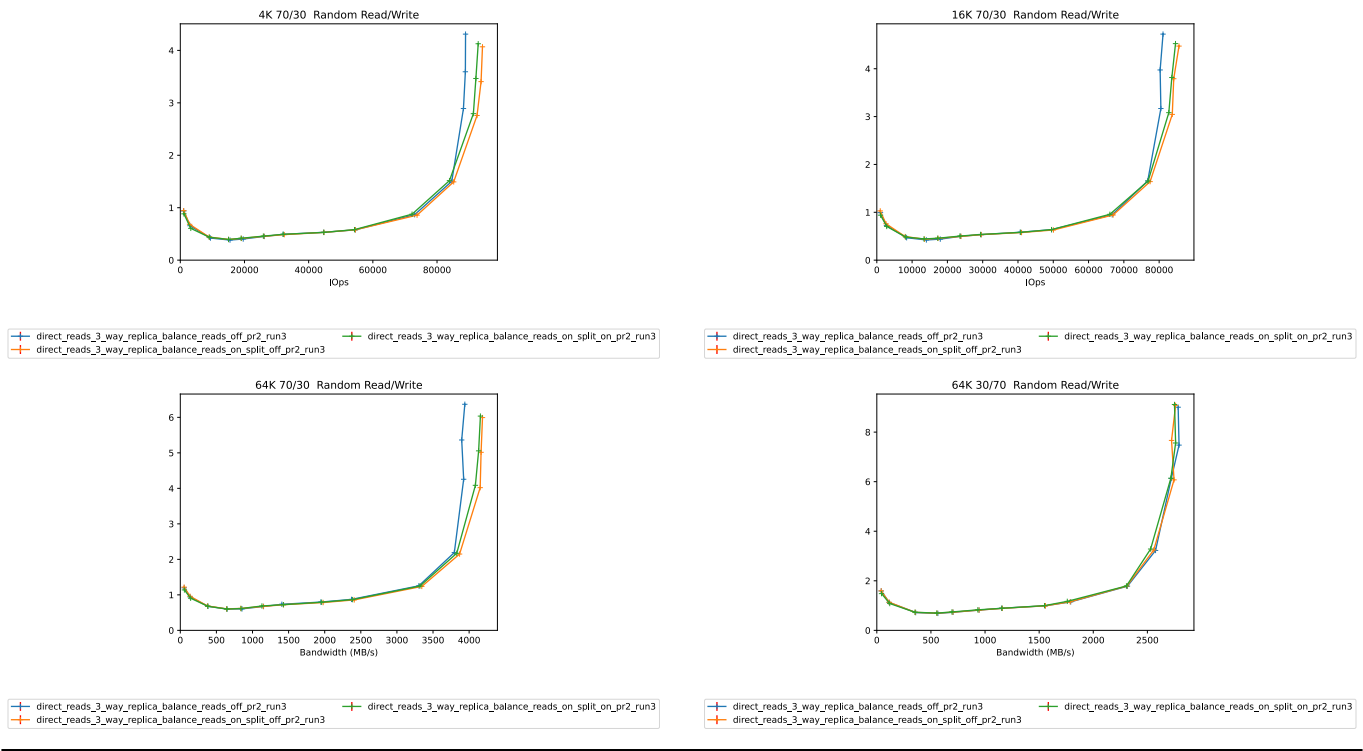


Random Read





Random Read/Write



Configuration yaml files

Only yaml files that differ by more than 20 lines from the yaml file for the baseline directory will be added here in addition to the baseline yaml

results

```

librbd fio:
  cmd_path: /usr/local/bin/fio2
  create_report: true
  fio_out_format: json
  log_avg_msec: 100
  log_bw: true
  log_iops: true
  log_lat: true
  norandommap: true
  osd_ra:
    - 4096
  poolname: rbd_replicated
  prefill:
    blocksize: 64k
    numjobs: 1
  procs_per_volume:
    - 1
  ramp: 30
  rbdname: cbt-librbd fio
  time: 90
  time_based: true
  use_existing_volumes: true
  vol_size: 1000
  volumes_per_client:
    - 16
  wait_pgautoscaler_timeout: 10
  workloads:
    128k randomread:
      jobname: randread
      mode: randread
      numjobs:
        - 1
      op_size: 131072
      total_iodepth:
        - 1
        - 2
        - 3
        - 4
        - 8
        - 12
        - 16
        - 20
        - 24
        - 28
        - 32
        - 64
        - 128
        - 256
        - 384
    128k sequentialread:
      jobname: seqread
      mode: read
      numjobs:
        - 1
      op_size: 131072
      total_iodepth:
        - 1
        - 2
        - 3
        - 4
        - 8

```

```

- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 256
- 384
16kmixread70:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 16384
  rwmixread: 70
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
16krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 16384
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
16ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 16384
  total_iodepth:
    - 1
    - 2
    - 4

```

```

- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
1Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 1048576
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 256
    - 384
1Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 1048576
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 256
    - 384
256krandomread:
  jobname: randomread
  mode: randread
  numjobs:
    - 1
  op_size: 262144
  total_iodepth:
    - 1

```

```

- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
256ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 262144
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
2Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 2097152
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 192
2Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 2097152
  total_iodepth:
    - 1

```

```

- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192
32krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 32768
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
32ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 32768
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
384krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 393216
  total_iodepth:
    - 1

```

```

- 2
- 3
- 4
- 5
- 8
- 16
- 24
- 32
- 64
- 128
- 160
- 256
- 512
384ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 393216
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 5
    - 8
    - 16
    - 24
    - 32
    - 64
    - 128
    - 160
    - 256
    - 512
4Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 4194304
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 192
4Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 4194304
  total_iodepth:
    - 1

```

- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192

4kmixread70:

- jobname: mixread
- mode: randrw
- numjobs:
- 1
- op_size: 4096
- rwmixread: 70
- total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

4krandomread:

- jobname: randread
- mode: randread
- numjobs:
- 1
- op_size: 4096
- total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

4ksequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 4096
- total_iodepth:

```

- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
512krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 524288
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 24
    - 32
    - 40
    - 48
    - 64
    - 128
    - 256
    - 512
512ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 524288
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 24
    - 32
    - 40
    - 48
    - 64
    - 128
    - 256
    - 512
64kmixread30:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 65536
  rwmixread: 30

```

```

total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
64kmixread70:
  jobname: mixread
  mode: randrw
  numjobs:
  - 1
  op_size: 65536
  rwmixread: 70
  total_iodepth:
  - 1
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
  - 320
  - 384
64krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 65536
  total_iodepth:
  - 1
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
  - 320
  - 384
64ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
  - 1

```

```

  op_size: 65536
  total_iodepth:
  - 1
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
  - 320
  - 384
768krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 786432
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
  - 24
  - 32
  - 40
  - 48
  - 64
  - 128
  - 256
  - 512
768ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 786432
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 5
  - 8
  - 16
  - 24
  - 32
  - 64
  - 128
  - 160
  - 256
  - 512
8krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1

```

```

      op_size: 8192
      total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
    8ksequentialread:
      jobname: seqread
      mode: read
      numjobs:
      - 1
      op_size: 8192
      total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
    precondition:
      jobname: precond1rw
      mode: randwrite
      monitor: false
      numjobs:
      - 1
      op_size: 65536
      time: 600
      total_iodepth:
      - 16
  cluster:
    archive_dir: /tmp/cbt
    ceph_mgr_cmd: /usr/bin/ceph-mgr
    ceph_mon_cmd: /usr/bin/ceph-mon
    ceph_osd_cmd: /usr/bin/ceph-osd
    ceph_run_cmd: /usr/bin/ceph-run
    ceph_cmd: /usr/bin/ceph
    clients:
    - --- server1 ---
    clusterid: ceph
    conf_file: /cbt/ceph.conf.4x1x1.fs
    fs: xfs
    head: --- server1 ---
    iterations: 1
    mgrs:
    --- server1 ---:

```

```

    a: null
mkfs_opts: -f -i size=2048
mons:
  --- server1 ---:
    a: --- IP Address ---:6789
mount_opts: -o inode64,noatime,logbsize=256k
osds:
- --- server1 ---
osds_per_node: 6
pdsh_ssh_args: -a -x -l%u %h
rados_cmd: /usr/bin/rados
rbd_cmd: /usr/bin/rbd
tmp_dir: /tmp/cbt
use_existing: true
user: cjames
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}

```